

■ Fact-Check Report

Source Document: Facts_Correct_False_Inaccurate.pdf
Report Generated: 2026-06-13 20:33:12

Summary Statistics

Total Claims	Verified	Inaccurate	False
5	2	0	3

Detailed Verification Results

Claim #1: ■ VERIFIED

Claim: The Earth revolves around the Sun.

Type: TECHNICAL

Page: 1

Confidence: 95%

Analysis: All provided sources unequivocally state and explain that the Earth revolves around the Sun. Sources 1, 2, 3, and 4 explicitly use terms like 'revolves around the Sun' or 'orbits the Sun', detailing the elliptical nature of the orbit, its duration, and the gravitational forces involved. Source 5, while a Reddit discussion, also confirms this premise while discussing orbital nuances.

Sources:

1. Earth's Rotation & Revolution Around the Sun Explained - Britannica - <https://www.britannica.com/video/Earth-rotation-axis-revolution-Sun/-159380>
2. Earth's orbit - Wikipedia - https://en.wikipedia.org/wiki/Earth%27s_orbit
3. Chapter 2: Reference Systems - NASA Science - <https://science.nasa.gov/learn/basics-of-space-flight/chapter2-1>

Claim #2: ■ VERIFIED

Claim: Water boils at 100C at standard atmospheric pressure.

Type: NUMERICAL

Page: 1

Confidence: 95%

Analysis: All provided sources consistently state that water boils at 100°C at standard atmospheric pressure, commonly defined as 1 atmosphere (1 atm or 101.325 kPa) at sea level. Some sources note a more precise value of 99.97°C, but 100°C is the widely accepted and historically defined temperature for this condition. The distinction with the IUPAC-recommended standard (99.61°C at 100 kPa) does

not invalidate the claim, which refers to 'standard atmospheric pressure' (1 atm).

Sources:

1. Boiling point - Wikipedia - https://en.wikipedia.org/wiki/Boiling_point
2. Boiling & Atmospheric Pressure - <https://terpconnect.umd.edu/~wbreslyn/chemistry/pressure/atmo-pressure.html>
3. Compound Interest: What Temperature Does Water Boil At? Boiling Point & Elevation - <https://www.compoundchem.com/2016/03/22/boiling-point>

Claim #3: ■ FALSE

Claim: Humans can breathe normally underwater without equipment.

Type: TECHNICAL

Page: 1

Confidence: 100%

Analysis: All provided sources consistently state that humans cannot breathe normally underwater without equipment. Source 1 explicitly says, 'Humans cannot breathe normally underwater without equipment; specialized gear like scuba tanks or rebreathers is needed.' Sources 4 and 5 discuss freediving, which involves holding one's breath for extended periods, not 'breathing normally.' Source 2 mentions innovative underwater breathing apparatus, but these are still 'equipment' and are not yet widely available or considered 'normal' breathing. Source 3 explains the physiological difficulties of breathing through even a simple tube underwater due to pressure differences, further underscoring that normal breathing without equipment is not possible.

Correct Information: Humans cannot breathe normally underwater without equipment. Without specialized gear, humans must hold their breath to survive underwater, a practice known as freediving.

Sources:

1. 3 Underwater Breathing Apparatus Innovations - <https://www.dresseldivers.com/blog/underwater-breathing-apparatus-innovations>
2. How far underwater could you breath using a hose or pipe (at 1 atmosphere) before the pressure becomes too much for your lungs to handle? : r/askscience - https://www.reddit.com/r/askscience/comments/3ap7dk/how_far_underwater_could_you_breath_using_a_hose
3. How Deep Can A Human Dive Without Scuba Gear? | Koox Diving - <https://www.kooxdiving.com/en/how-deep-can-a-human-dive-without-scuba-gear>

Claim #4: ■ FALSE

Claim: The Moon produces its own light.

Type: TECHNICAL

Page: 1

Confidence: 100%

Analysis: All provided sources, including official NASA content, clearly state that the Moon does not produce its own light. What we perceive as "moonlight" is actually sunlight reflecting off the Moon's surface. The Sun is the only object in our solar system that produces its own light.

Correct Information: The Moon reflects sunlight; it does not produce its own light.

Sources:

1. Moonlight - NASA Science - <https://science.nasa.gov/moon/moonlight>
2. Moon Phases - NASA Science - <https://science.nasa.gov/moon/moon-phases>
3. Does The Moon Produce Its Own Light? Lets Find Out... - <https://www.youtube.com/watch?v=M8L3wfxSmgw>

Claim #5: ■ FALSE

Claim: The Great Wall of China is visible from space with the naked eye.

Type: TECHNICAL

Page: 1

Confidence: 100%

Analysis: All provided sources unequivocally state that the Great Wall of China is not visible from space with the naked eye. Astronauts and space agencies (like NASA) confirm that no man-made structures are visible from space without magnification. While the wall is very long, its narrow width makes it essentially invisible to the human eye from orbit. It can sometimes be photographed or observed with powerful cameras or telescopes, but not with the unaided eye.

Correct Information: The Great Wall of China is not visible from space with the naked eye.

Sources:

1. What makes the Great Wall of China the only man-made object ... - <https://www.wtamu.edu/~cbaird/sq/2012/12/11/what-makes-the-great-wall-of-china-the-only-man-made-object-visible-from-space>
2. Google Earth , Structures and Anomalies | I think most of us know it is a myth that the Great Wall of China is the only man-made structure that is visible from space | Facebook - <https://www.facebook.com/groups/253550249143357/posts/1142773493554357>
3. TIL the Great Wall of China is not actually visible from space, its just ... - https://www.reddit.com/r/todayilearned/comments/1kg3uwe/til_the_great_wall_of_china_is_not_actually